

Sealed-seam suppression in Froggatt–Nielsen-type models: two theorems, an obstruction by conjunction, and a discrete neutrino test

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preprint, not peer reviewed, version 2026-06-11 (substantially rebuilt after hostile review; the withdrawn claims of the earlier draft are listed in Section 7 and stay withdrawn).

Abstract

Froggatt–Nielsen (FN) mechanisms generate fermion mass hierarchies as powers of a small parameter λ , one insertion per unit of flavor charge, with λ fitted. We study an FN-type setting in which the suppression is *pinned by a theorem*: for sealed two-outcome records — rank-one POVM-type letter pairs with weights p and $1 - p$ in a stationary representation — (i) the transition norm is $\sqrt{p(1-p)} |\langle u, v \rangle| \leq \sqrt{p(1-p)}$, and (ii) a misaligned sealed pair cannot support a stationary state (top transfer eigenvalue $\frac{1}{2}[1 + \sqrt{1 - 4p(1-p)(1 - |c|^2)}] < 1$ strictly), so stationarity forces saturation: $\|S\| = \sqrt{p(1-p)}$ exactly, with the seam phase pure rephasing gauge. An exact selection rule follows in any algebra graded by a family charge whose only charged sealed generator is the seam: amplitudes vanish unless the seam count balances the charge difference (receipts: 0/27,000 violations; bounds saturated to ten digits). The paper’s central *negative* result is an **obstruction by conjunction**, found when the hostile review of an earlier draft demolished its seesaw realization and we completed the demolition: for *integer* right-handed charges, Majorana diagonals carry even powers of λ and an exhaustive Dirac-consistent scan over rule-allowed textures (charges in $\{0..3\}^3$, unit $\pm$ coefficients) never reaches the half-integer-rung spectrum (zero exact hits; best deviation 0.014; receipted); for *half-odd* right-handed charges the Majorana side can reach the rungs exactly, but the corresponding Dirac column carries half-integer charge, balances no integer seam count, and vanishes identically — rank collapse; and the textbook cancellation in any case removes RH charges from the light masses when the Dirac sector is charged consistently. The surviving realization places half-unit charges on the lepton doublets (Weinberg-operator dressing), where the same selection rule yields a **discrete menu** for the light spectrum at unit coefficients (four full-rank members; two already excluded by data): the rung point $S = \sqrt{\varepsilon_{\text{eff}}/(1 + \varepsilon_{\text{eff}})} = 0.172747$ (three insertions) and the cheaper texture $S = 0.167880$ (two insertions), with $\varepsilon_{\text{eff}} = \varepsilon(1 - \varepsilon)$ and $\varepsilon = 0.031768636446582$ a constant whose self-contained pedigree is the mathematical companion [C0]. Against JUNO’s first data ($S = 0.17283 \pm 0.00167$, our combination of JUNO’s Δm_{21}^2 with the global-fit Δm_{31}^2) the data-live members sit at -0.05σ and -3.0σ ; they are separated by 7.3σ at JUNO-final precision, so the menu is adjudicable this decade, and the minimality principle the program uses elsewhere favors — on current data — the *worse* member, which we state rather than hide. What this paper does **not** deliver, withdrawn from an earlier draft and listed with reasons: mixing-angle

and CP predictions (they rested on an unlisted measure-zero orthogonality assumption), the claim that the mechanism’s flavor symmetry uniquely ”predicts its own mediator,” and any charged-fermion statement. One quantitative input is irreducible and stated in one line: the seam weight is the companion theory’s first-order marginality, $p = \varepsilon$ — a referent scan shows it underivable from the obvious fixed-point quantities, and it carries the aggravating fact that the first-order law it comes from has its sign overturned at second order at the defining weight. The look-elsewhere cost of the dressed value is counted (six of twelve natural one-factor forms fit today’s central value; three are mutually inseparable even at JUNO-final), and the claim chronology relative to JUNO’s first release is disclosed in full.

1. The Seam Theorem

Work in the representation every reflection-positive stationary process admits (companion [C2], Thm 2.1): Hilbert space, PSD letters, $T = \sum_x A_x \preceq \mathbf{1}$, stationary unit vector $T\omega = \omega$.

Definition (S-atom premise, named). A *sealed two-outcome seam* is a letter pair of rank one: $A_0 = p u u^\dagger$, $A_1 = (1 - p) v v^\dagger$, unit u, v .

Lemma 1 (norm). $\sqrt{A_0}\sqrt{A_1} = \sqrt{p(1-p)} \langle u, v \rangle u v^\dagger$, so $\|\sqrt{A_0}\sqrt{A_1}\| = \sqrt{p(1-p)} |\langle u, v \rangle| \leq \sqrt{p(1-p)}$, with equality iff the poles align.

Lemma 2 (stationarity gate). On the seam span, with $c = \langle u, v \rangle$ and $d = 4p(1-p)(1-|c|^2)$: $\lambda_{\max}(A_0 + A_1) = \frac{1}{2}[1 + \sqrt{1-d}]$, equal to 1 iff $|c| = 1$; the exact gap is $\frac{1}{2}[1 - \sqrt{1-d}] \geq p(1-p)(1-|c|^2)/2$, strictly positive for misaligned poles. (An earlier draft’s abstract quoted the lower bound as the exact gap; corrected.)

Theorem 1 (stationarity forces saturation). If the seam block acts invariantly at stationarity (premise: decoupling) and the stationary state overlaps it, then alignment is forced and the transition norm is exactly $\sqrt{p(1-p)}$. At alignment the letters commute: the extremal seam is classical, and the factor is the two-pole weight of a definite record. The seam phase ($u \mapsto e^{i\varphi}u$) changes no observable — the rephasing structure mass terms have always had.

Receipts: norm identity to 2.8×10^{-16} (200 random complex seams, $d = 2.5$); eigenvalue closed form to 3.3×10^{-16} .

2. The selection rule

Theorem 2. In an algebra carrying a graded charge Q with every sealed generator Q -neutral except one seam S shifting Q by one unit, $\langle x'|W|x \rangle = 0$ for every word whose seam count fails to balance $x' - x$ (grade decomposition), and — since neutral letters are contractions — balanced amplitudes obey $|\langle x'|W|x \rangle| \leq \|S\|^{|x'-x|}$ (submultiplicativity; an inequality, with saturation a property of the specific model, exhibited in the receipts).

Receipts: 0 violations in 27,000 matrix elements; maximal one- and two-unit amplitudes 0.175384 and 0.030759, saturating $\sqrt{\varepsilon_{\text{eff}}}$ and ε_{eff} to ten digits in the explicit model.

Scope correction (withdrawn corollary). An earlier draft claimed ”oscillations require the seam — the mechanism predicts its own mediator.” That conflated two gradings: if the Dirac sector is charged consistently, the claim’s premise breaks the spectrum (Section 3); if the Dirac

sector is outside the graded algebra, sectors connect without the seam and the corollary is vacuous. Withdrawn. What survives is the unglamorous statement: *within the graded (sealed) sector*, charge steps cost seam insertions.

3. The obstruction by conjunction (central result)

The natural realization — FN charges on the right-handed seals, M_R hierarchical, Dirac sector neutral — **fails twice over**, and we record both failures as the paper’s main content:

(a) The textbook cancellation. If the Dirac couplings are charged consistently with the grading, the FN suppressions cancel identically in the seesaw, $m_\nu = v^2 Y M_R^{-1} Y^\top$ — the well-known result that FN charges on singlet neutrinos drop out of the light masses (see e.g. [2]); and half-integer RH charges make the Dirac column amplitudes exactly zero under a unit-shifting seam, collapsing the rank. An earlier draft evaded this by declaring the Dirac sector “neutral” while keeping the selection rule — an inconsistency the review exposed.

(b) Parity, scoped, plus the scan. Majorana bilinears $\nu_i^c \nu_j^c$ carry total charge $x_i + x_j$. For **integer** charges the diagonals carry even powers of λ — a parity statement that by itself constrains only diagonal-dominant textures (off-diagonal-dominant blocks can carry odd-power eigenvalue magnitudes), so the integer-charge obstruction is completed by *exhaustive enumeration*: over charge vectors in $\{0..3\}^3$ with unit $\pm$ coefficients on every rule-allowed entry, **zero** textures reach the rung spectrum. The closest approach — metric pinned as the max over the two independent ratio deviations $|m_1/m_3 - \varepsilon_{\text{eff}}|$ and $|m_2/m_3 - \sqrt{\varepsilon_{\text{eff}}}|$, **computed on the seesaw-inverted (light) spectrum**: the charges sit on M_R and the rung target is for the light masses, so $m_i \propto 1/\mu_i$ with μ_i the eigenvalue magnitudes of the scanned texture (equivalently, the two components are $|\mu_1/\mu_3 - \varepsilon_{\text{eff}}|$ and $|\mu_1/\mu_2 - \sqrt{\varepsilon_{\text{eff}}}|$) — is 0.014, with the **witness published** so the figure is checkable from one matrix: $q = (3, 2, 3)$, coefficient pattern $\{11:+1, 12:+1, 13:+1, 22:-1, 23:+1, 33:0\}$ on magnitudes $\lambda^{q_i+q_j}$, component deviations (0.0144, 0.0092). (Applying the metric to the heavy spectrum *uninverted* gives (0.0144, 0.0770) on this same witness; an earlier draft attributed that 0.077 to “a less complete scan variant” — it is in fact the witness’s own uninverted second component, and the inversion convention was unstated — the same class of convention bug a previous revision fixed for the charge box, caught in review and now pinned.) The zero-exact-hits statement is robust under the pinned convention: a full independent rerun of the $\{0..3\}^3 \times \{0, \pm 1\}^6$ scan confirms zero exact hits, best deviation 0.0144. Scope honesty: the charge box $\{0..3\}^3$ is a choice; entries beyond it carry magnitudes $\leq \lambda^7 \approx 5 \times 10^{-6}$ and push textures further from the target, but “exhaustive” means exhaustive-within-the-box. For **half-odd** charges, parity does *not* obstruct the Majorana side — $q = (1, \frac{1}{2}, 0)$ makes $M_R = \text{diag}(\lambda^2, \lambda, 1)$ rule-allowed and exactly on the rungs — but clause (a) kills the realization: the $q = \frac{1}{2}$ Dirac column carries half-integer charge, balances no integer seam count, and is exactly zero, collapsing the light-mass rank. **The obstruction is the conjunction of (a) and (b)-plus-scan**, and we state it as such; an earlier draft presented (b) as a standalone parity theorem (false for half the charge lattice) and left the scan’s spectrum-inversion convention unstated (the source of the misattributed 0.077 above) — both corrected here, with the scan scope and conventions now pinned.

The surviving realization. Half-unit charges on the lepton *doublets*, $q_L = (1, \frac{1}{2}, 0)$, acting on the Weinberg operator $(L_i H)(L_j H)$ with entry charges $q_i + q_j$ and magnitudes $\lambda^{q_i+q_j}$ at integer insertions ($\lambda = \sqrt{\varepsilon_{\text{eff}}}$): allowed entries are (11), (22), (33), (13); (12) and (23) are parity-forbidden.

This realization is *selected, not chosen*, by two facts: (i) **integer** doublet charges on the Weinberg operator are excluded by the same exhaustive enumeration run on the *direct* spectrum (no seesaw inversion here) — zero exact hits over the full $\{0..3\}^3 \times \{0, \pm 1\}^6$ box, 30,624 full-rank non-degenerate textures, best deviation 0.0770, witness in the receipts; and (ii) $q_L = (1, \frac{1}{2}, 0)$ is the *unique* normalized assignment realizing the rung diagonal: $\text{diag}(\lambda^{2q_1}, \lambda^{2q_2}, \lambda^{2q_3}) = \text{diag}(\lambda^2, \lambda, 1)$ forces $2q_1 = 2$, $2q_2 = 1$, with the common shift fixed by $q_3 = 0$ (a shift rescales the spectrum, not its ratios). Convention, pinned: entries are 0 or ± 1 times the allowed magnitude; full rank and a nondegenerate ordering ($m_1 < m_2 < m_3$, all nonzero) required. The exhaustive menu then has exactly **four** members (a previous revision said three, missing the quasi-degenerate fourth — caught in review; for a paper whose method is exhaustive enumeration, we print the complete list):

texture	insertions	spectrum ratios	S
<code>diag(1^2, 1, 1)</code>	3	<code>(e_eff, sqrt(e_eff), 1)</code>	0.172747
<code>m13 + m22 + m33</code>	2	<code>(0.0290, 0.1703, 1)</code>	0.167880
<code>all four, sgn(m11 m33) = -1</code>	4	<code>(0.0581, 0.1704, 1)</code>	0.160500
<code>m11 + m22 + m13, m33 = 0</code>	4	<code>(0.839, 0.916, 1)</code>	0.675516
(the third and fourth are already excluded by data - at 7.4 and ~300 sigma respectively - and are listed as falsified members, not hidden; an earlier draft printed 0.169993 for the second member, a wrong closed-form shortcut - the correct value follows from the printed ratios themselves: $S^2 =$ $m1/(1+2m1)$, $m1 = (\text{sqrt}(1+4*e_eff)-1)/2$, receipted)			

4. The two-point test

Against JUNO's first data — $S = 0.17283 \pm 0.00167$, **our combination** of JUNO's $\Delta m_{21}^2 = (7.50 \pm 0.12) \times 10^{-5}$ [3] with the global-fit $\Delta m_{31}^2 = (2.511 \pm 0.027) \times 10^{-3}$ [6], not a JUNO-only number:

member	S	now	notes
rung point	0.172747	-0.05 s	(reference)
cheap texture	0.167880	-3.0 s	7.3 s from the rung point at JUNO-final
third member	0.160500	-7.4 s	already excluded;
fourth member	0.675516	excluded	listed, not hidden
(undressed point for comparison	0.175472	+1.6 s	the registered claim of the companion note [VII])

The two data-live members are separated by 7.3σ at JUNO-final precision ($\sigma_S^{\text{final}} \approx 0.00067$: the projected sub-percent splittings, $\sigma_{21} \approx 0.6\%$ and $\sigma_{31} \approx 0.5\%$ [5], propagated as half the quadrature sum of the relative errors times S): the menu is adjudicable this decade. Pre-assigned outcome bands at JUNO-final, all branches covered: data within 2σ of 0.172747 — rung point survives, cheap texture dead; within 2σ of 0.167880 — the reverse, and minimality scores a sharp win; **within $2\sigma_{\text{final}}$ of 0.175472 — both menu members die while [VII]'s undressed point survives**; anywhere else — the whole construction is dead and we say so. Stated against interest: the program's own minimality principle (fewest insertions), applied as elsewhere in the corpus, favors the **cheap texture** — currently the *worst-fitting live member*

at -3.0σ . If JUNO-final confirms today’s central value, minimality-as-selector fails here and we will say so.

What the menu does *not* include: mixing angles and CP phases. The allowed Weinberg textures are too sparse to generate the observed mixing from the neutrino sector alone (the (12), (23) entries are parity-forbidden), so the PMNS matrix must come dominantly from the charged-lepton sector — the earlier draft’s θ_{23} and δ_{CP} ”predictions” are **withdrawn** (Section 7). One genuine charged-sector *constraint* the realization does impose, stated since review caught us denying it: $q_{L_2} = \frac{1}{2}$ makes the muon Yukawa row $\bar{L}_2 e_R H$ carry half-integer charge, which vanishes unless the right-handed charged-lepton charges compensate modulo 1 — half-odd e_{R_2} is an **additional named assumption** (a massless muon is not negotiable). So the mechanism says nothing about charged *masses* but does constrain charged *charges*; the abstract’s withdrawal item is scoped accordingly.

5. The one-line input, with its aggravating fact

The mechanism’s single quantitative input: the seam weight is the companion theory’s first-order marginality, $p = \varepsilon = 3\kappa - 1$, with κ in closed form ([C0]; no Standard-Model input anywhere in the chain). A pre-registered referent scan (ten exact fixed-point candidates, 25 digits) found no independent quantity equal to ε : the identification is not derived, and we state the aggravating fact a referee located: ε is the coefficient of a *first-order* law whose sign prediction fails at the defining weight $w = 3$ (first-order -0.0051 vs exact $+0.0084$; [C0] itself warns against extrapolating its truncations). The input is therefore: *the seam weight is a specific coefficient of an expansion that is not trustworthy at the weight that motivates it*. We keep the input because the resulting point sits at -0.05σ and the falsification machinery is cheap; we decline to dress it up.

6. Chronology, trials, and look-elsewhere

Disclosed in full. (i) The *value* $\varepsilon_{\text{eff}} = \varepsilon(1 - \varepsilon)$ was noticed **after** JUNO’s first release; the program’s registered claim is the *undressed* point (0.175472 at exact ε ; see [VII] and its precision correction), which that data disfavors at 1.6σ . (ii) The look-elsewhere cost was counted before this mechanism existed. The twelve forms, enumerated (base $\varepsilon \cdot f$): $f \in \{1 - \varepsilon, 1/(1 + \varepsilon), e^{-\varepsilon}, (1 - \varepsilon)^2, 1 - 2\varepsilon, 1 - \varepsilon/2, 1/(1 + 2\varepsilon), 1 - \varepsilon^2, \cos \varepsilon, 1 - \kappa\varepsilon, 1 - \theta\varepsilon, 1 - \eta\varepsilon\}$: six fit the current central value within 1σ ; three (including $\varepsilon_{\text{eff}} = \varepsilon(1 - \varepsilon)$) are mutually inseparable at 0.2σ even at JUNO-final — so the data can confirm or kill the one-factor *class*, never select $\varepsilon(1 - \varepsilon)$ within it, and a theorem pinning any cluster member would have looked equally prophetic. The Seam Theorem’s evidential weight is therefore *zero until its premises are tested elsewhere*; its value is that it converts a free parameter into a falsifiable conjunction. (iii) No claim in this paper is ”registered” in any externally verifiable sense until the program’s named deposit exists; the internal registration documents carry their own dating and say the same.

7. Withdrawn claims (kept visible)

1. ” $\sin^2 \theta_{23} = 1/2$ exactly” — rested on requiring an *orthogonal* neutral Dirac matrix, a measure-zero assumption with no symmetry behind it, which also smuggled CP conser-

vation (real orthogonal) and silently fixed θ_{12}, θ_{13} to excluded values. Withdrawn as a mechanism prediction; what remains is a texture observation in the program’s corpus (its registration document carries the same withdrawal).

2. “ $J = 0, \delta_{CP} \in \{0, \pi\}$ ” — the phase-gauge argument covers the seam (Majorana) sector only; the Dirac and charged-lepton phases are untouched. Withdrawn.
3. “Oscillations require the seam” — grading-scope conflation (Section 2). Withdrawn.
4. The discrete $m_{\beta\beta}$ menu — computed with undisclosed measured mixing angles as inputs; meaningless once (1) is withdrawn. Withdrawn.
5. “Exact rungs for any neutral Dirac sector” — exactness was an artifact of the orthogonality fiat; the honest exact statement is the Weinberg-realization menu of Section 3.
6. All charged-fermion *mass* statements — the earlier draft quantified charged-sector “gaps” with an unreproducible metric; no such number survives. (Not withdrawn: the charge-mod-1 compensation *constraint* of Section 4, which the realization genuinely imposes and which carries its own named assumption.)

8. Relation to Froggatt–Nielsen models

The mechanism class is FN [1]. Prior art on *derived* suppression exists and is engaged per our standing commitment: in anomalous- $U(1)$ (Green–Schwarz) FN models the parameter is computed from the Fayet–Iliopoulos term, $\lambda \approx 0.17\text{--}0.22$ (Binétruy–Ramond; Dudas–Pokorski–Savoy; Ibáñez–Ross [4]) — a range that, we note with appropriate discomfort, contains $\sqrt{\varepsilon_{\text{eff}}} = 0.175$. What is different here: the suppression is pinned by a stationarity theorem in a record formalism rather than by an anomaly coefficient, the charge’s selection rule is exact rather than order-of-magnitude, and the realization question (this paper’s obstruction and menu) is answered by exhaustive enumeration at unit coefficients. What is *not* different: like every FN model, the $O(1)$ -coefficient freedom is the mechanism’s soft underbelly, and our unit- coefficient discipline is a named choice, not a derivation.

Reproducibility

Fixed-seed scripts regenerate every number (the seam lemmas, the selection-rule receipts, the parity-obstruction scan, the Weinberg menu, the prediction table); bit-identical reruns; code and canonical outputs accompany the submission.

References

- [1] C. D. Froggatt, H. B. Nielsen, *Nucl. Phys. B* **147** (1979) 277.
- [2] G. Altarelli, F. Feruglio, Models of neutrino masses and mixings, *New J. Phys.* **6** (2004) 106 — including the cancellation of singlet-neutrino FN charges in the seesaw.
- [3] JUNO collaboration, arXiv:2511.14593; *Nature* **654** (2026) 343 — first measurement, $\Delta m_{21}^2 = (7.50 \pm 0.12) \times 10^{-5} \text{ eV}^2$.

[4] P. Binétruy, P. Ramond, *Phys. Lett. B* **350** (1995) 49; E. Dudas, S. Pokorski, C. A. Savoy, *Phys. Lett. B* **356** (1995) 45; L. E. Ibáñez, G. G. Ross, *Phys. Lett. B* **332** (1994) 100 — anomalous- $U(1)$ FN with FI-derived λ .

[5] JUNO collaboration, JUNO physics and detector, *Prog. Part. Nucl. Phys.* **123** (2022) 103927 — projected sub-percent splitting precisions (an earlier draft cited the DUNE TDR and Hyper-Kamiokande design report here; those experiments do not measure Δm_{21}^2 at sub-percent — misattribution, corrected).

[6] I. Esteban et al., NuFIT — global-fit Δm_{31}^2 input; version sensitivity applies to the hybrid S above.

Companions: [C0] the fixed-point theory (the constant’s pedigree); [C2] the representation theorem; [VII] the registered undressed point and its correction record (the two claims are mutually exclusive and cross-declared).